

Epidemic Modeling

Epidemic - an event involving a spreading disease that ultimately infects a majority of a population

Epidemic modeling mathematically describes the spread of a disease through a population. This helps scientists gain further insight into the effects of a disease's spread through a population.

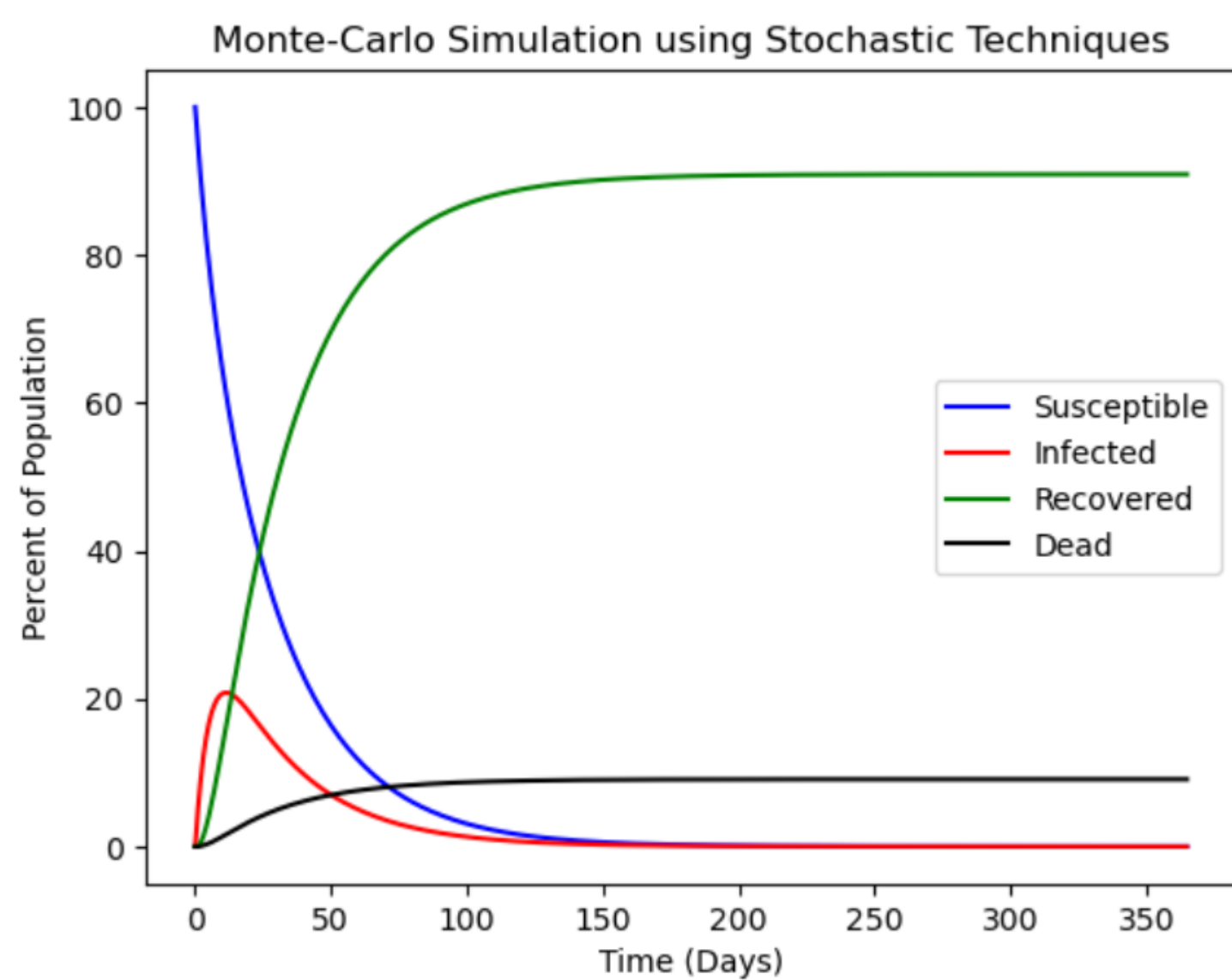
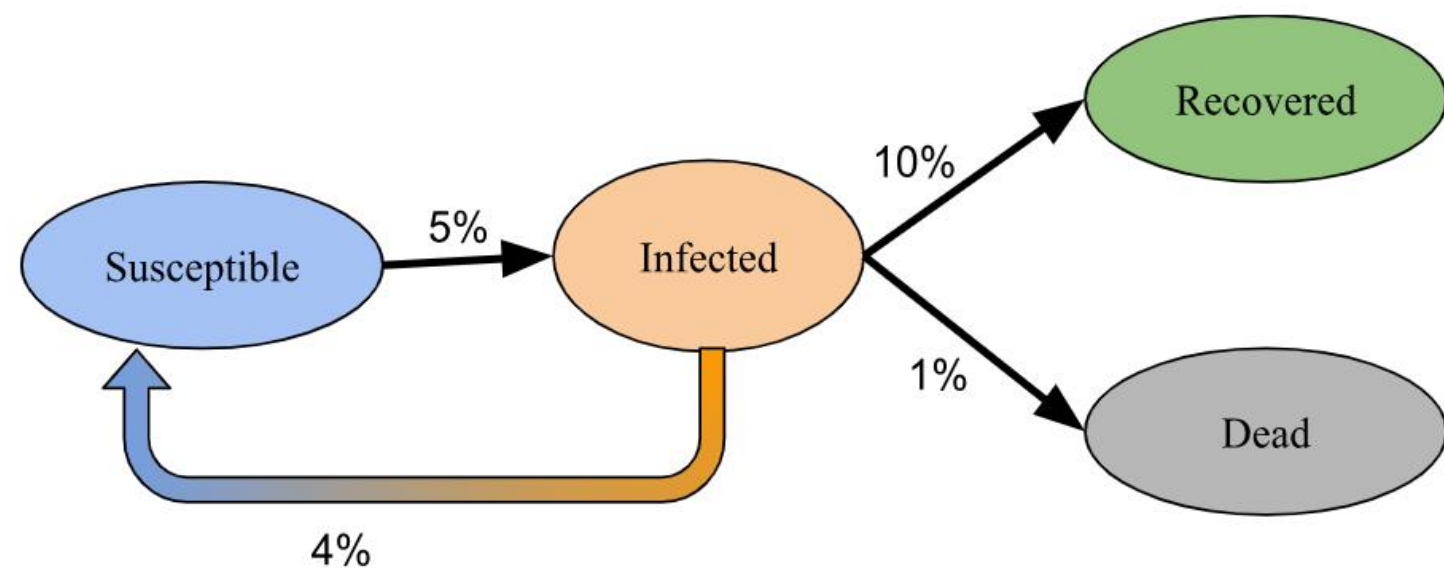
Introduction

We sought to achieve two goals:

- Examine the comparative effect of vaccination and social distancing on the maximum number of infections from a given disease
- This work provides insights into the dynamics of a disease and may inform public health policies and responses to epidemics
- Examine the difference in the rate of infections on populations categorized by the type of work they perform
- Categorizing the population based on work they perform creates a relation between the number of interactions and number of infections, which describes the effectiveness of a disease's ability to spread.
- We divided the population into blue collar, white collar, and stay-at-home groups

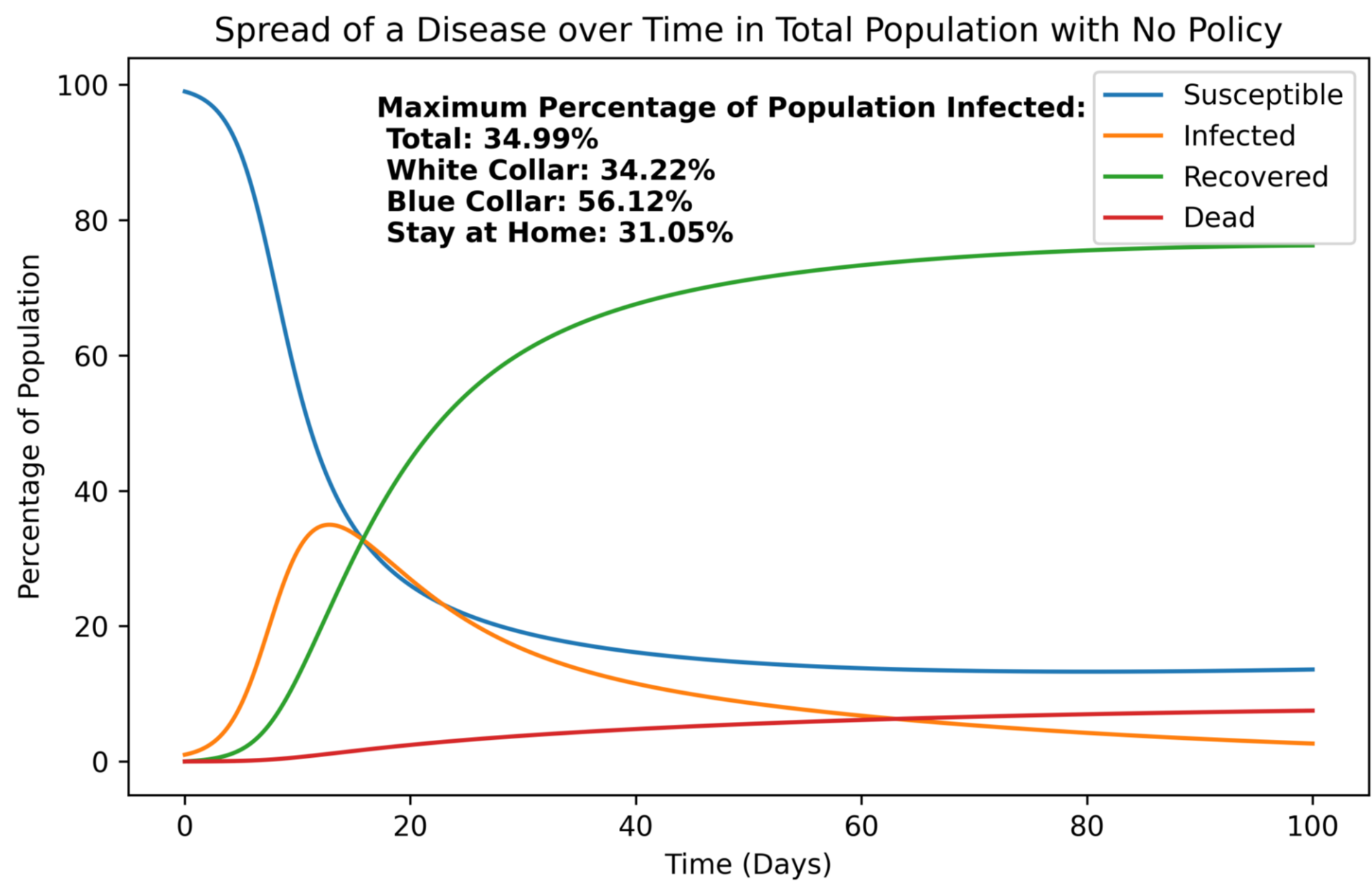
Stochastic Model

Transition of Populations between Status Groups Based on Given Probabilities

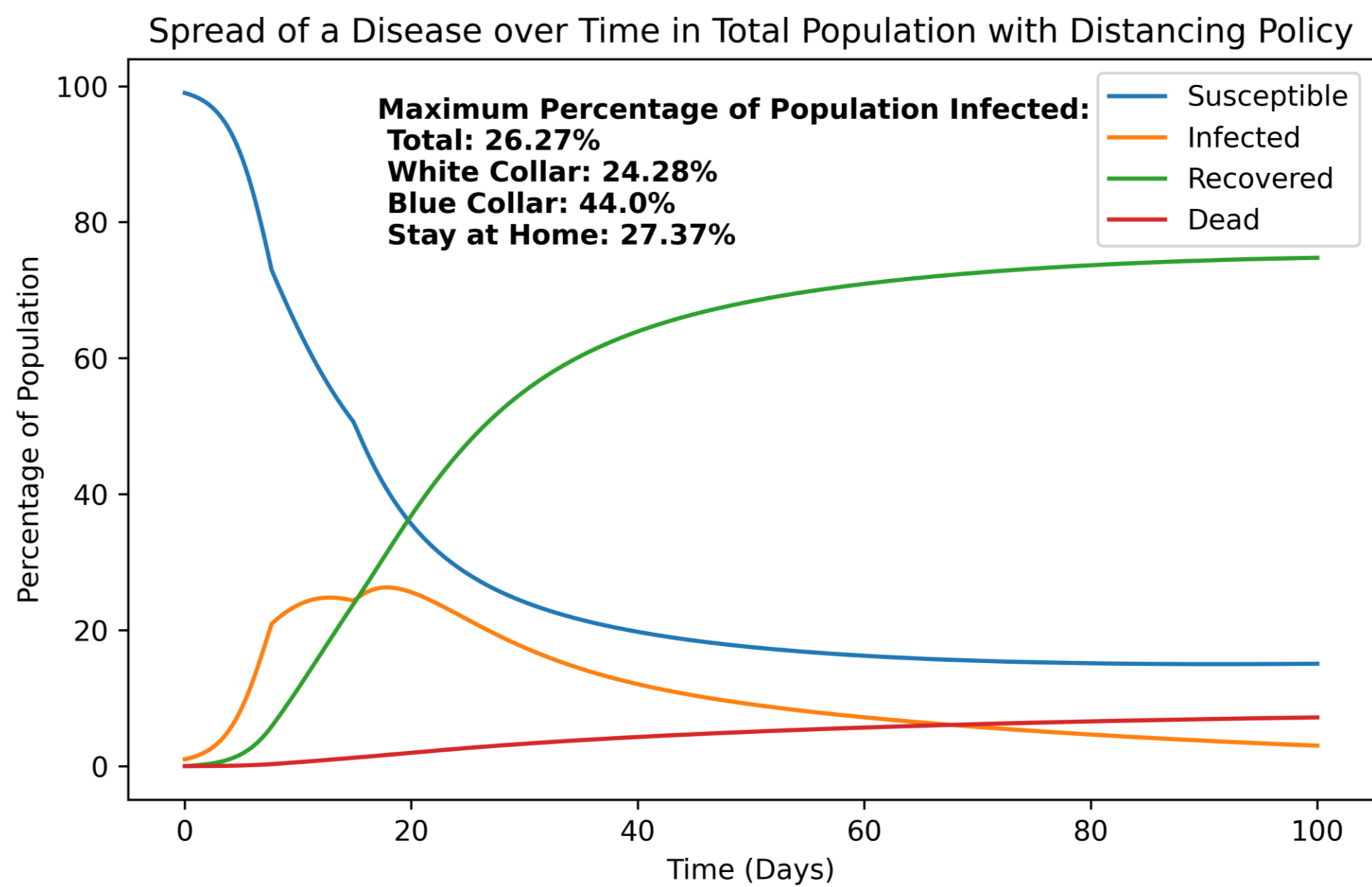


Results of Various Policy Choices

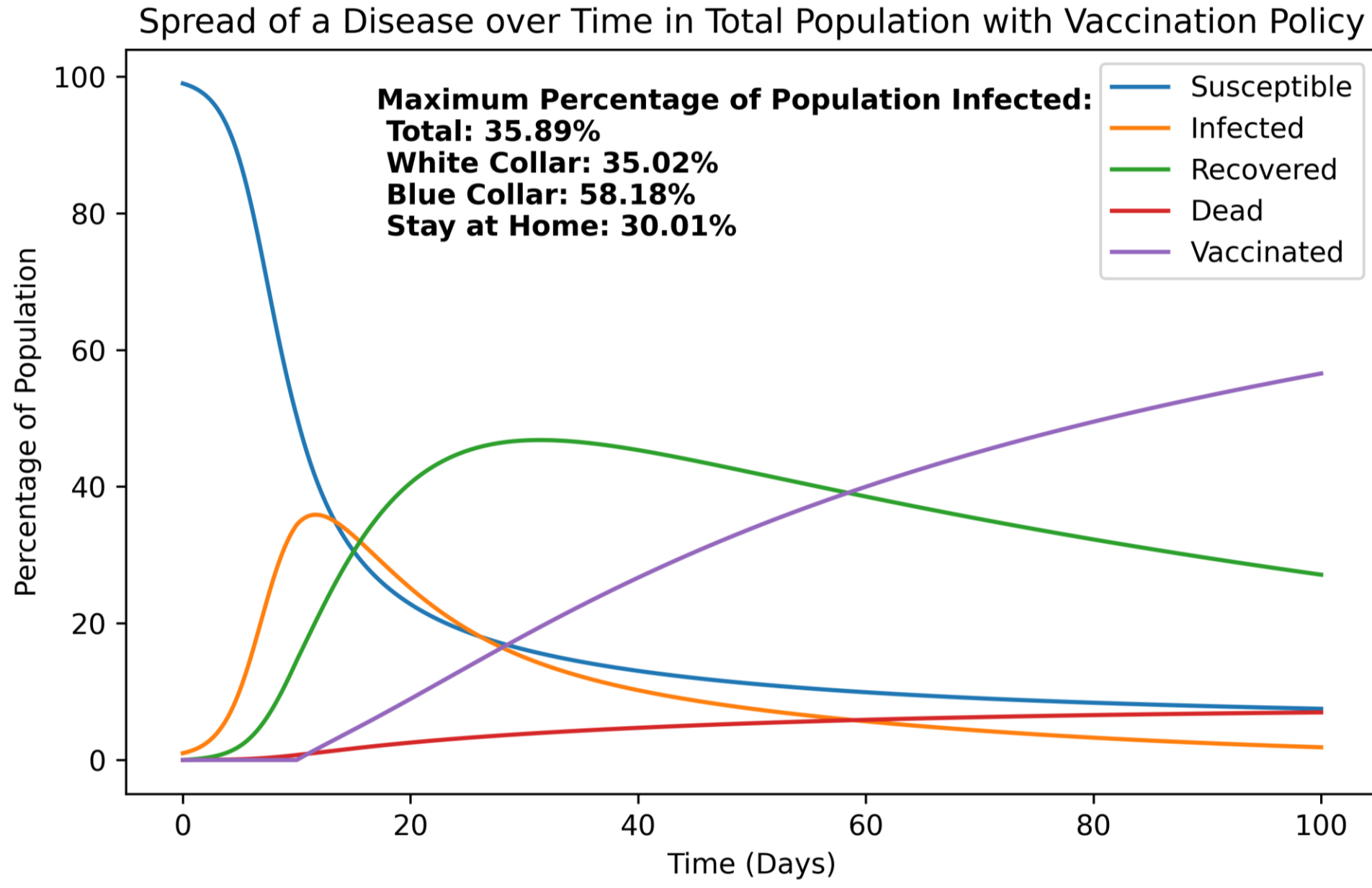
None



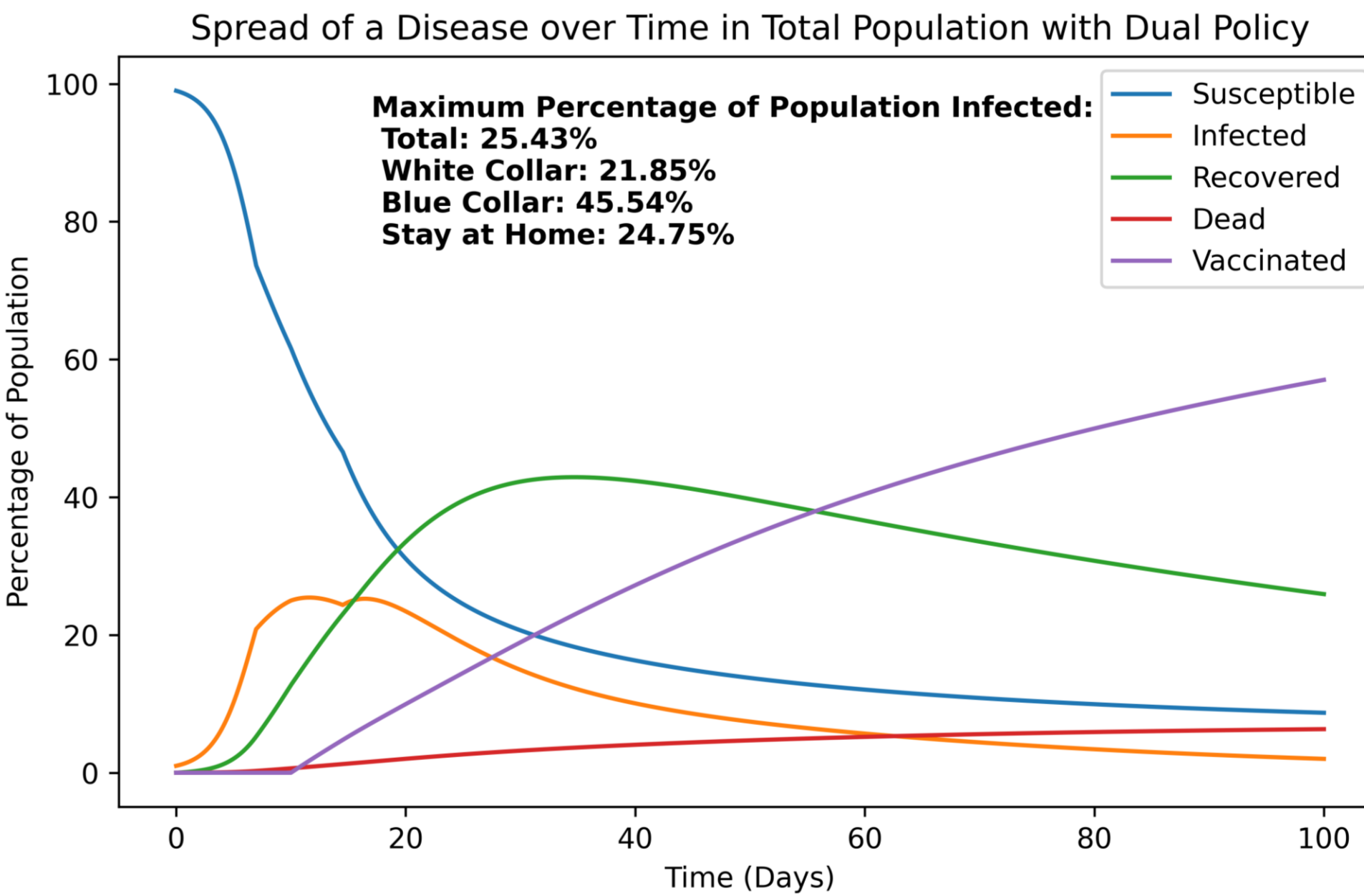
Social Distancing



Vaccination



Vaccination and Social Distancing



Results

- The **most effective policy** at reducing the spread of a disease was the combination between vaccination and social distancing policies.
 - This approach successfully shortens the length of the epidemic and helps achieve herd immunity.
- The social distancing policy had the **largest stand-alone impact** on reducing total number of infected people.
 - For a short epidemic, this policy is most effective.
 - However, keeping people separate from one another could lead to various other health issues.
- The vaccination policy is the most beneficial to **achieving herd immunity** (which prevents likelihood of further epidemics).
 - However, we realize vaccination policies have many obstacles to their implementation.
 - The vaccine policy also requires a longer time period to be effective.

Further Exploration

- Investigate the impact of social distancing on mental health and immune system strength
- Explore the impact of temporary versus long-term social distancing policies
- Investigate the likelihood of vaccination based on type of work done

References

- Martignoni, M. M., Raulo, A., Linkovski, O., & Kolodny, O. (2024). SIR+ models: accounting for interaction-dependent disease susceptibility in the planning of public health interventions. *Scientific Reports*, 14(1), 12908. <https://doi.org/10.1038/s41598-024-63008-9>
- This research was supported in part by an appointment to the RENEW Pathway Summer Schools Program at the Oak Ridge National Laboratory, sponsored by the U.S. Department of Energy and administered by the Oak Ridge Institute for Science and Education.

Modeling Techniques

Agent-Based Monte Carlo Techniques – models based on a set of “agents,” or people, that make decisions and act according to a set of rules

SIR Models – models using a set of coupled ordinary non-linear differential equations to describe the spread of a disease through a population, mathematically describe disease spread

Agent-Based Monte-Carlo Models

Pros

- I. Includes more descriptors
- II. Includes more localized data

Cons

- I. Computational complexity – it takes a lot of power to run the simulation

SIR Models

Pros

- I. Computationally efficient – less power required to run on a computer

Cons

- I. Cannot include as many descriptors
- II. Cannot include localized data

Coupled Differential Equations

- For a set of populations x, y, and z, the set of equations has the form:

$$\frac{dS_x}{dt} = -\alpha_x S_x - \beta_{xx} S_x I_x - \beta_{xy} S_x I_y - \beta_{xz} S_x I_z$$

$$\frac{dI_x}{dt} = \alpha_x S_x - \gamma_x I_x - \delta_x I_x + \beta_{xx} S_x I_x + \beta_{xy} S_x I_y + \beta_{xz} S_x I_z$$

$$\frac{dR_x}{dt} = \gamma_x I_x$$

$$\frac{dD_x}{dt} = \delta_x I_x$$

Where:

S_x is the number of susceptible people in class x

I_x is the number of infected people in class x

R_x is the number of recovered people in class x

D_x is the number of dead people in class x

α_x is the probability of infection for class x (in per day)

β_{xx} is the number of interactions between people in class x (in per day)

β_{xy} is the number of interactions between people in class x and y (in per day)

β_{xz} is the number of interactions between people in class x and z (in per day)

γ_x is the probability of recovery for people in class x (in per day)

δ_x is the probability of death for people in class x (in per day)